ABIRAMI R

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PROFILE SUMMARY

Developer & Machine Learning Engineer with expertise in **Python, Java, ReactJS, SQL,** Skilled in building **end-to-end applications** and developing **ML models using TensorFlow, Keras, PyTorch, and Scikit-learn**, with a strong foundation in **Exploratory data analysis.**

SKILLS

Analytical tools: Spice simulation, Git, Microsoft excel.
Languages known: Python, HTML, CSS, Javascript, Java.
Frameworks / technologies: ReactJS, Bootstrap, Tailwind CSS, Flask, FastAPI, TensorFlow, Keras, PyTorch, Scikit-learn, NLP, Pandas, NumPy, MySQL, MongoDB, Microcontrollers (Arduino, ESP32), Embedded System, REST APIs, fine-tuning LLMs and Hugging Face Transformers.

EXPERIENCE

CSIR – National Aerospace Laboratories - Bengaluru, Karnataka. Aug 2025-Present

ML Project Associate

- Developed an acoustic source localization framework during flight test instrumentation, utilizing MFC/PZT rosette sensors integrated with the **KAM-500 DAQ system** for high-fidelity, real-time data acquisition.
- Implemented wavelet-based Lamb wave amplitude extraction and advanced signal processing to analyze structural responses under dynamic test conditions.
- Estimated wave direction of arrival through dynamic **strain tensor analysis** and multi-rosette triangulation for accurate damage localization.
- Designed a machine learning pipeline for MAT module for real-time processing, converting sensor data into actionable insights during live testing scenarios.
- Transformed processed directional data into physics-informed boundary conditions for a **PINN-based inverse localization model.**
- Achieved ~98% localization accuracy with **~10 mm MSE,** outperforming traditional RNN-based approaches in complex, anisotropic aerospace structures.

CONTRIBUTIONS

Google Summer of Code 2025, Google. Apr 2025

Student developer program – Contributor (Accord project)

- Migrated template playground's legacy CSS codebase to Tailwind CSS to enhance scalability, maintainability, and development speed.
- Refactored core UI components like navbar, buttons, forms, and modals using Tailwind's utility-first classes.
- Reduced CSS bundle size and eliminated redundant styles, **improving overall performance and load times by ~25%**.
- Collaborated with maintainers to get PRs merged, including hover interaction fixes and consistent styling patterns.

PROJECTS & RESEARCH

SatDecay PINNs - Python – Git - Pytorch. 2025

- Built a hybrid physics + ML framework to improve **satellite re-entry prediction** for Low Earth Orbit objects.
- Combined **SGP4 orbital propagation** with data-driven error correction using historical TLE and decay datasets.
- Engineered Gradient Boosting and **PINN models to capture atmospheric drag** variability and nonlinear orbital decay.
- Delivered a computationally efficient pipeline suitable for real-time debris tracking and operational forecasting systems.

Iot-bedside control panel Arduino - Embedded systems - GIT. 2025

- Designed and developed a **smart IoT-based bedside control panel** equipped with sensors for temperature, gyroscope, and heart rate monitoring, improving monitoring coverage **from 60% to 100%**.
- Implemented **automatic SMS alerts** via a GSM module to notify caregivers in case of abnormal heart rate, reducing average response time **from 10 minutes to under 6 minutes (70% to 90%).**
- Click here : Iot-control-panel

EDUCATION

B.E. in Electronics and Communication Engineering. 2021-2025
DMI College Of Engineering, Chennai

- CGPA: 77.59%

+2 PCM. 2020-2021
St. Columban's A.I.HR.SEC School, Chennai

- GPA: 84%

PROFESSIONAL AFFILIATIONS

- Member, **International Association of Engineers.**
- Citizen Scientist**, NASA, US.**